

E-Commerce Architecture

Proposal & Comparison

Clothing Brand Digital Storefront

This document presents every architecture option considered for your e-commerce project — from simple managed solutions to fully custom stacks. Each option is evaluated on cost, traffic capacity, ease of management, and technical complexity.

Prepared by your development team

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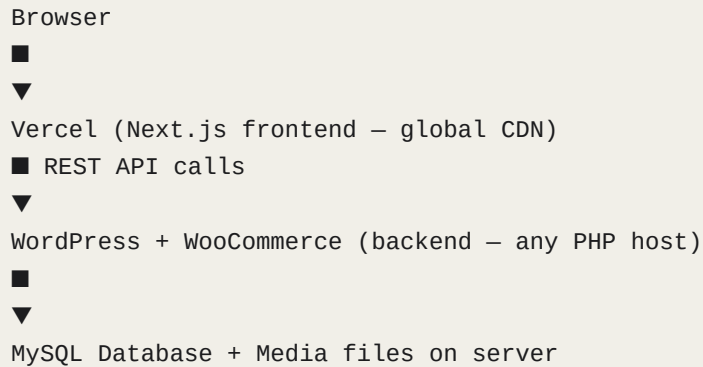
1	WordPress + WooCommerce + Vercel	Managed CMS backend with Next.js frontend
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ARCHITECTURE 1

WordPress + WooCommerce + Vercel

The classic approach: WordPress manages products and orders; Next.js delivers the storefront.

HOW IT WORKS



TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
\$12/mo host	~40 users	~20k/month	Too small — MySQL starves PHP
\$24/mo host	~200 users	~80k/month	Good for a new brand
\$48/mo host	~500 users	~200k/month	Handles campaigns and flash sales

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan covers a new store
Shared PHP hosting	\$12–48/mo	Hostinger, SiteGround, WP Engine
Domain	~€15/year	Registered separately
WooCommerce plugins	\$0–150/year	Stripe, PDF invoices, etc.
Stripe payment processing	1.5% + €0.25	Per transaction (EU cards)
Total (starting)	~\$17–53/mo	Excluding transaction fees

PROS & CONS

Advantages	Limitations
✓ Non-technical staff can manage products and orders immediately	✗ WordPress + plugins slow the API down significantly
✓ 300+ payment plugins — one-click Stripe, PayPal, Klarna, Mollie	✗ Security patches and plugin updates require ongoing maintenance
✓ WooCommerce order dashboard is polished and client-friendly	✗ Limited control — plugin conflicts are common
✓ Huge plugin ecosystem — reviews, loyalty, abandoned cart emails	✗ Performance ceiling lower than a custom API
✓ No custom development for admin features	✗ Not truly headless — some WooCommerce logic leaks to frontend

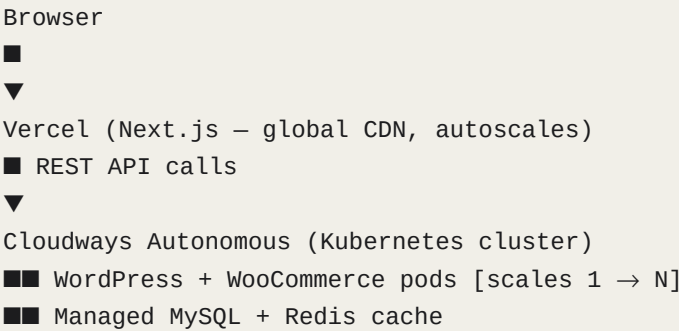
Verdict: Best for teams that want zero custom code and a polished admin out of the box. Accept a performance and maintenance cost in return.

ARCHITECTURE 2

Vercel + Cloudways Autonomous

WordPress on a Kubernetes-backed managed platform that autoscales automatically.

HOW IT WORKS



TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
Entry plan	~300 users	~120k/month	Autoscales within plan limits
Growth plan	~1,000 users	~400k/month	Kubernetes adds pods on spikes
Scale plan	Unlimited	Millions	True horizontal scaling

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan
Cloudways Autonomous	\$35–120/mo	Entry to Growth plan
Domain	~€15/year	Registered separately
Stripe processing	1.5% + €0.25	Per transaction
Total	~\$35–120/mo	Scales with your traffic

PROS & CONS

Advantages	Limitations
✓ True autoscaling — survives a viral Instagram post	✗ More expensive than a single Droplet at low traffic
✓ Managed infrastructure — no server maintenance	✗ Still carries WordPress performance overhead
✓ WordPress admin is client-friendly out of the box	✗ Less control than a self-managed server
✓ Cloudflare Enterprise CDN built in	✗ Plugin updates still required
✓ Daily backups and one-click restores included	✗ Headless WooCommerce has some compatibility quirks

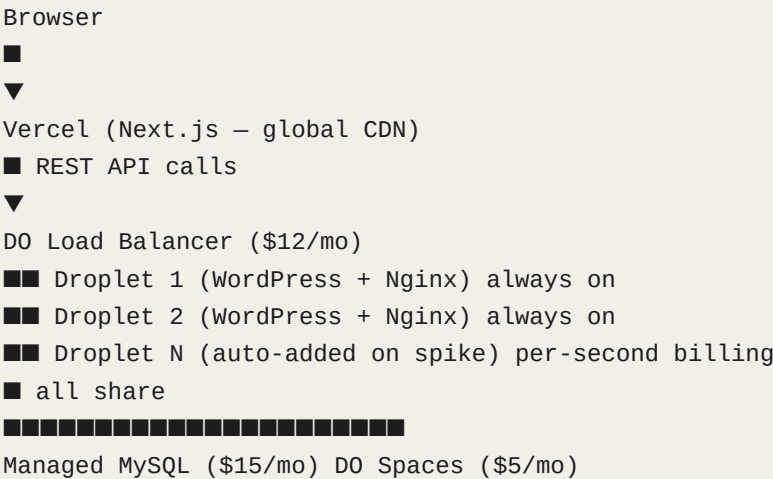
Verdict: Best choice if you want WordPress's ease of use AND protection against traffic spikes, and don't mind paying a premium for it.

ARCHITECTURE 3

Vercel + DigitalOcean Autoscale Pool

Multiple identical servers behind a load balancer, sharing one database.

HOW IT WORKS



TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
2× \$24 Droplets	~400 users	~160k/month	Baseline — always running
4× \$24 Droplets	~800 users	~320k/month	Auto-added during spikes
N× Droplets	Linear scale	Unlimited	Pay per second for extra nodes

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	
DO Load Balancer	\$12/mo	Always on
2× Base Droplets	\$48/mo	WordPress + Nginx
Managed MySQL	\$15/mo	Shared by all Droplets
DO Spaces + CDN	\$5/mo	Shared media storage

Extra Droplets	Cents/spike	Per-second billing
Total (normal)	~\$80/mo	Rises during traffic spikes

PROS & CONS

Advantages	Limitations
✓ True horizontal autoscaling on DigitalOcean	✗ More complex to set up than a single Droplet
✓ No single point of failure — load balancer routes around crashes	✗ Requires shared media storage (DO Spaces) — extra config
✓ Per-second billing makes spike costs minimal	✗ Still carries WordPress overhead on each node
✓ Full control over server configuration	✗ Managed MySQL is a separate cost even at low traffic
✓ Cheaper than Cloudways at the same traffic level	✗ Database becomes bottleneck at very high scale

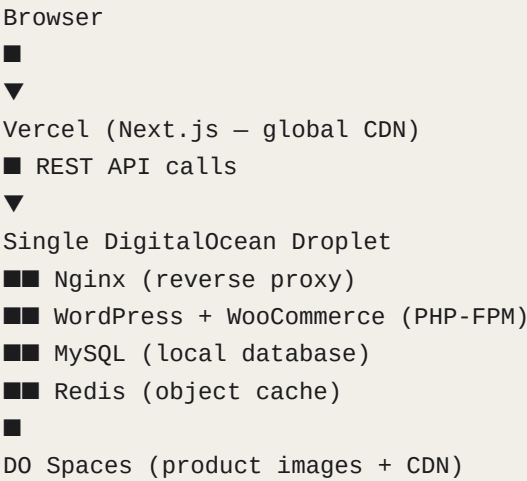
Verdict: Best for brands that want self-managed infrastructure with real autoscaling. More work to set up; pays off at medium to high traffic.

ARCHITECTURE 4

Vercel + Single DigitalOcean Droplet

Everything on one server: WordPress, MySQL, and Redis. Simple and affordable.

HOW IT WORKS



TRAFFIC CAPACITY (WordPress stack)

Droplet size	Concurrent users	Monthly visitors	Notes
\$12/mo 1 vCPU 1GB	~40 users	~20k/mo	Too small for MySQL + PHP together
\$24/mo 2 vCPU 4GB	~200 users	~80k/mo	Sweet spot for a new brand
\$48/mo 4 vCPU 8GB	~500 users	~200k/mo	Handles campaigns and flash sales
\$96/mo 8 vCPU 32GB	~1,500 users	~600k/mo	Single server limit — consider splitting

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	

DO Droplet \$24	\$24/mo	2 vCPU, 4GB RAM — recommended start
DO Spaces + CDN	\$5/mo	Product images
Stripe processing	1.5% + €0.25	Per transaction
Total	~\$29/mo	Cheapest production-grade option

PROS & CONS

Advantages	Limitations
✓ Cheapest production setup at ~\$29/mo	✗ No autoscaling — a traffic spike can take the site down
✓ Simple — one server to manage	✗ Single point of failure — one crash, everything is offline
✓ Easily resized vertically in the DO dashboard	✗ WordPress performance overhead still applies
✓ Clear upgrade path to autoscale pool when needed	✗ MySQL and PHP compete for the same RAM
✓ Full control over configuration	✗ Must manually resize to handle growth

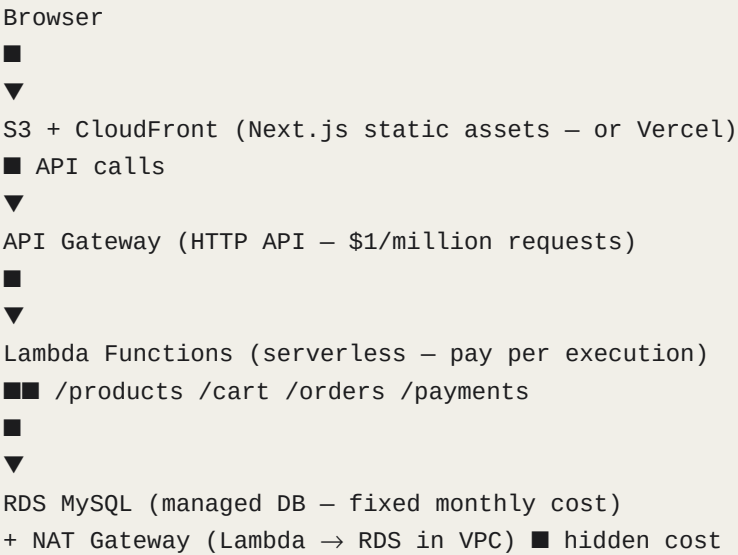
Verdict: Best starting point for a new brand. Low cost, easy to manage. Upgrade to Architecture 3 when you need autoscaling.

ARCHITECTURE 5

Vercel + AWS (S3 + Lambda + RDS)

Fully serverless: pay only for what you use, scales to millions automatically.

HOW IT WORKS



TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
Free tier	~1M req/mo	Lambda free tier	No cost for compute at low scale
db.t3.micro	Low-medium	~40k visitors/mo	\$22/mo fixed even at zero traffic
db.t3.small	Medium	~200k visitors/mo	\$30/mo — grows with product catalog
Unlimited	Infinite	Autoscales with Lambda	DB is the only bottleneck

COSTS

Service	Cost/mo	Notes
S3 + CloudFront	~\$5/mo	Frontend assets
API Gateway	~\$5–12/mo	HTTP API, pay per request

Lambda compute	~\$3–8/mo	Nearly free at small scale
RDS MySQL t3.small	~\$30/mo	Fixed — runs 24/7
NAT Gateway	~\$33/mo	Required for Lambda → RDS. Hidden cost!
CloudWatch logs	~\$5–10/mo	Monitoring
Total	~\$90–120/mo	Much higher than expected due to NAT

PROS & CONS

Advantages	Limitations
✓ Infinite autoscaling — Lambda handles any traffic spike	✗ NAT Gateway adds ~\$33/mo — often exceeds Lambda cost
✓ Pay per request — zero cost at zero traffic (Lambda)	✗ Most complex setup of all options considered
✓ AWS reliability — 99.99% uptime SLA	✗ RDS is a fixed cost even when the store is quiet
✓ Full control over every component	✗ Requires deep AWS knowledge to maintain
✓ Industry-standard — any developer knows this stack	✗ Cold starts add latency on Lambda first requests
	✗ Overkill for a single clothing brand store

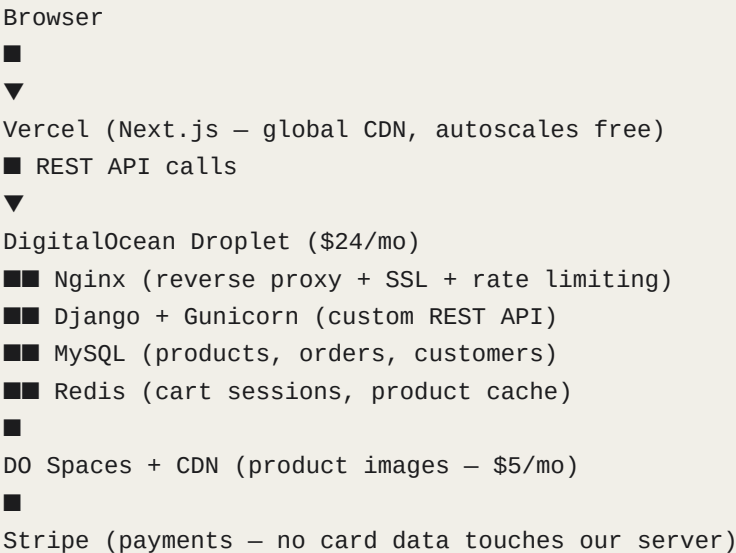
Verdict: Powerful but over-engineered for this project. The NAT Gateway alone costs more than our entire recommended stack. Only choose this if your team already runs AWS infrastructure.

ARCHITECTURE 6

Vercel + Django API + DigitalOcean

Custom Python API replaces WordPress entirely. Maximum performance, full control.

HOW IT WORKS



TRAFFIC CAPACITY (Django stack — 2–3× more than WordPress)

Droplet size	Concurrent users	Monthly visitors	Notes
\$12/mo 1 vCPU 1GB	~100 users	~40k/mo	Minimum viable — dev/staging only
\$24/mo 2 vCPU 4GB	~500 users	~200k/mo	Recommended start — sweet spot
\$48/mo 4 vCPU 8GB	~1,200 users	~500k/mo	Growing brand — handles flash sales
\$72/mo 8 vCPU 16GB	~2,500 users	~1M+/mo	Consider separate DB at this level

WHY DJANGO HANDLES MORE TRAFFIC ON SAME HARDWARE

Django (this stack)	WordPress (same Droplet)
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Idle RAM usage	~180 MB	~380 MB
Code on each request	Only your code runs	10–30 plugins execute
Product page with Redis	0 DB queries (cached)	3–8 DB queries
Checkout (uncacheable)	Same — hits DB live	Same — hits DB live

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan covers any new store
DO Droplet \$24	\$24/mo	Django + MySQL + Redis + Nginx
DO Spaces + CDN	\$5/mo	250GB product images + global CDN
Domain	~€15/year	.gr domain
dev.domain.gr	\$12/mo	Separate dev Droplet — optional
Stripe processing	1.5% + €0.25	Per transaction (EU cards)
Total (production)	~\$29/mo	Same cost as WordPress — 2.5× faster

WHAT THE CLIENT MANAGES DAY-TO-DAY

Task	How	Difficulty
Add a product (with sizes/colours)	Django admin — inline variant form	Easy after setup
Process an order	Change status → auto-emails customer	Very easy
Run a discount / sale	Create discount code in admin	Easy
Upload product images	Drag and drop in admin	Easy
View sales reports	Custom dashboard page in admin	Read-only — very easy
Issue a refund	Stripe dashboard (separate tab)	Easy
Bulk upload products via Excel	django-import-export plugin	Easy — one upload

PROS & CONS

Advantages	Limitations
✓ 2–3× more traffic than WordPress on identical hardware	✗ Requires 1–2 days of custom admin configuration
✓ No plugin tax — only your code runs on every request	✗ Admin is not as polished as WooCommerce out of the box
✓ Full control — any feature can be built exactly as needed	✗ Refunds go through Stripe dashboard separately
✓ Redis caches product pages — near-zero DB load when browsing	✗ Any new admin feature needs development time
✓ Custom admin can be tailored to the client's exact workflow	✗ More initial setup work than WordPress
✓ Same \$29/mo cost as the cheapest WordPress setup	
✓ Clear upgrade path — add autoscale pool when needed	

Verdict: OUR RECOMMENDATION. Same monthly cost as WordPress, 2.5× the traffic capacity, and a cleaner codebase. 1–2 days of admin customisation gives the client a comfortable, tailored experience. The right choice for a brand that wants to grow.

SUMMARY

All Architectures at a Glance

Use this table to compare every option side by side.

WordPress + Vercel (shared host)	\$17–53	80k–200k/ mo	No	★★★★★	Low	Starting out
Cloudways Autonomous + Vercel	\$35–120	400k+/mo	Yes	★★★★★	Low	Easy + scale
DO Autoscale Pool + Vercel	\$80+	Unlimited	Yes	★★★	Medium	Medium traffic
DO Single Droplet + WP + Vercel	\$29	80k–200k/ mo	No	★★★★★	Low	Cheapest start
AWS Lambda + RDS + Vercel	\$90–120	Unlimited	Yes	★★	High	AWS teams only
Django + DO Droplet + Vercel ♦	\$29	200k–500k /mo	No→Yes*	★★★★★	Medium	OUR PICK

* Django stack on single Droplet has no autoscale. Adding the DO autoscale pool later is a straightforward migration.

RECOMMENDATION

What We Recommend and Why

After evaluating all six architectures against your requirements — a clothing brand e-commerce site with non-technical staff, low admin overhead, good performance, and room to grow — we recommend:

Architecture 6

Vercel + Django API + DigitalOcean Droplet

- ❖ **Same price as the cheapest option**
\$29/mo — identical to a basic WordPress setup, but with 2.5× the traffic capacity.
- ❖ **Best performance per euro**
Django's idle memory is half of WordPress. Redis caches product pages so the database is barely touched during browsing. Your storefront will load faster.
- ❖ **Client-friendly admin**
1–2 days of customisation gives the brand team a tailored dashboard: inline product variants, order status with auto-emails, discount codes, sales reports, and bulk Excel product upload.
- ❖ **No plugin anxiety**
No WordPress plugins to update, no security patches from third-party authors, no plugin conflicts. Only your code runs.
- ❖ **Clear upgrade path**
When traffic grows, moving MySQL to a Managed Database costs \$15/mo extra. Adding the autoscale pool is a straightforward migration with no downtime.
- ❖ **Production-grade security built in**
JWT authentication, Stripe webhook signature verification, atomic stock decrement, rate limiting, HTTPS, and admin 2FA are all included.

SCALING ROADMAP

Launch	\$29/mo	Single Droplet + Django + Vercel. Handles up to 500 concurrent users.
Growth	\$44/mo	Add Managed MySQL (+\$15/mo). Frees Droplet RAM for more traffic.
Scale	\$80+/mo	Add DO Load Balancer + autoscale pool. Handles any traffic spike.

Enterprise

Custom

Multi-region deployment, read replicas, dedicated CDN contracts.